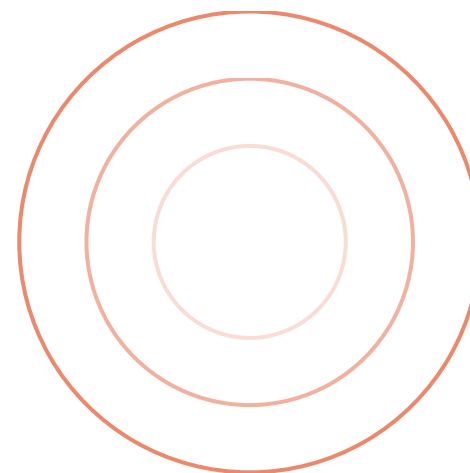


SERVERLESS ON CLOUDLET

From Knative to the Portal

What we built, and why.

Team review · September 2026



Four chapters

- 01 Foundations: serverless, Knative, the Operator
- 02 The API: contract, mTLS, two regions
- 03 Functions: buildpacks, kpack, build controller
- 04 Portal and cloudlet-apis: what comes next

01 → 04

fourteen questions, one platform

What is serverless?

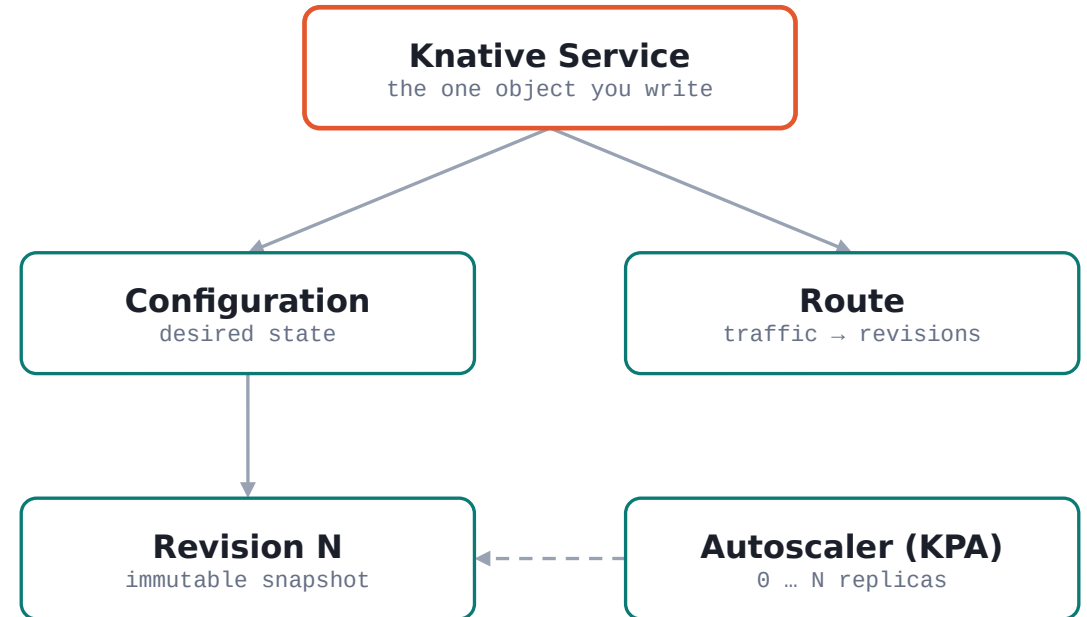
- You bring code, not servers
- Scale to zero when idle
- Scale out when traffic arrives
- Pay for requests, not hours
- Nothing to patch, nothing to plan



replicas follow traffic

What is Knative?

- Kubernetes building blocks for serverless
- Service owns Configuration and Route
- Every change is an immutable Revision
- KPA: concurrency in, replicas out
- Activator wakes revisions from zero
- DomainMapping: your own hostname



The OpenShift Serverless Operator

- Red Hat's supported Knative distribution
- One KnativeServing custom resource
- OLM installs it and upgrades it
- Kourier ingress, Routes created for you
- Catalog mirrors with oc-mirror

Knative Serving

Service · Revision · Route ·
DomainMapping

KnativeServing CR

one document

Serverless Operator

reconciles, upgrades

Operator Lifecycle Manager

Subscription → catalog

OpenShift

one cluster per region

Why the Operator

- Every alternative made us own more
- We never create a Route ourselves
- Airgap needs one supported mirror path

	UPSTREAM KNATIVE	SERVERLESS OPERATOR
Upgrades	YAML we maintain	OLM, from the catalog
Ingress	Run it ourselves	Kourier, managed
Routes, TLS	Hand-made, RBAC	Created for us
Airgap	Image by image	oc-mirror, once
Support	Community	Red Hat

02

The API

One HTTP call, a workload on every cluster.

One call, every cluster

- Functions from git, containers from images
- Validate now, answer 202, poll status
- Closed status vocabulary, published on /info
- FastAPI, Pydantic, cloudlet-apis
- One Helm chart, rendered per region

2

offerings

202

every write

0

kubectl for users

2

regions per deploy

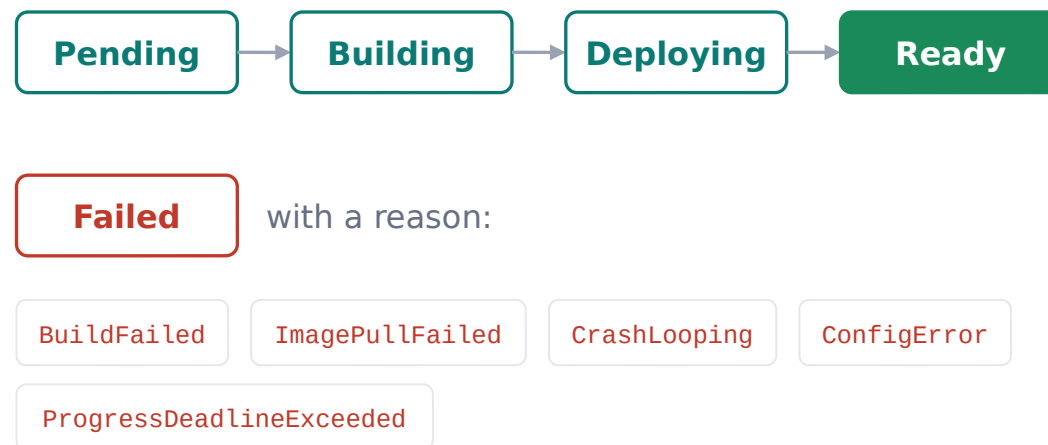
The endpoint structure

- 23 endpoints under one base path
- The group is in the path
- OIDC token or admin key
- Streams use short-lived tickets

```
/api/serverless/v1  
/groups/{group}/functions  
  POST · GET · PUT · DELETE  
  /{name}/build · /stats · /pods · /logs  
/groups/{group}/containers → /pull  
/functions/info · /containers/info  
/stream-tickets
```

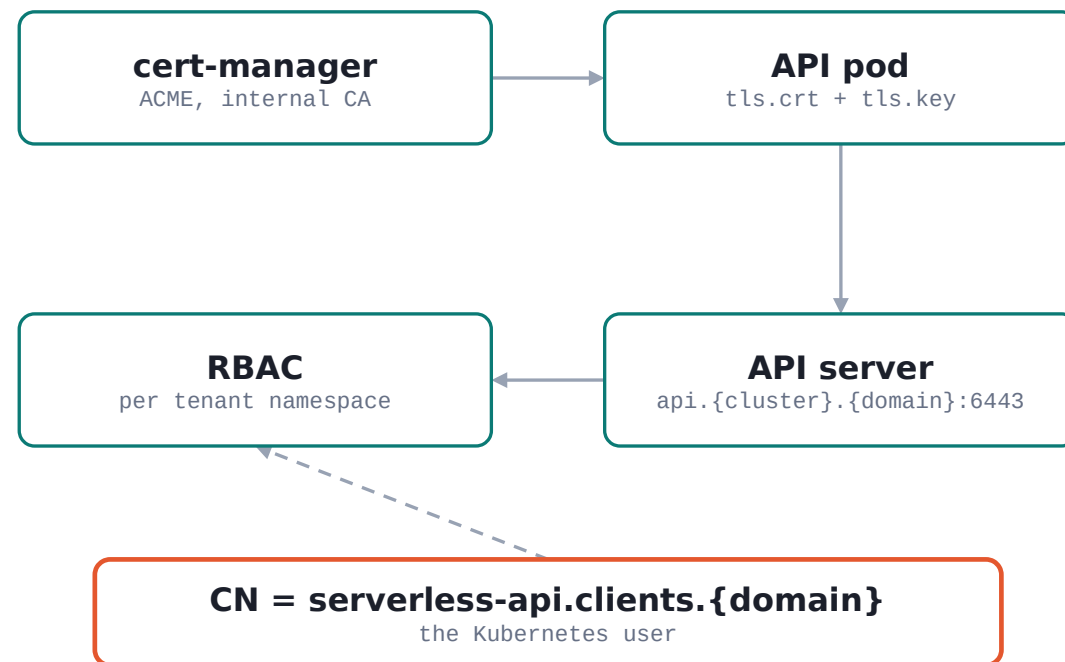
Anatomy of a workload

- Identity: name, hostname, regions
- Source: git repo or image
- Runtime: env, files, port, size
- Scaling: min, max, metric, target
- Response: same shape, secrets redacted



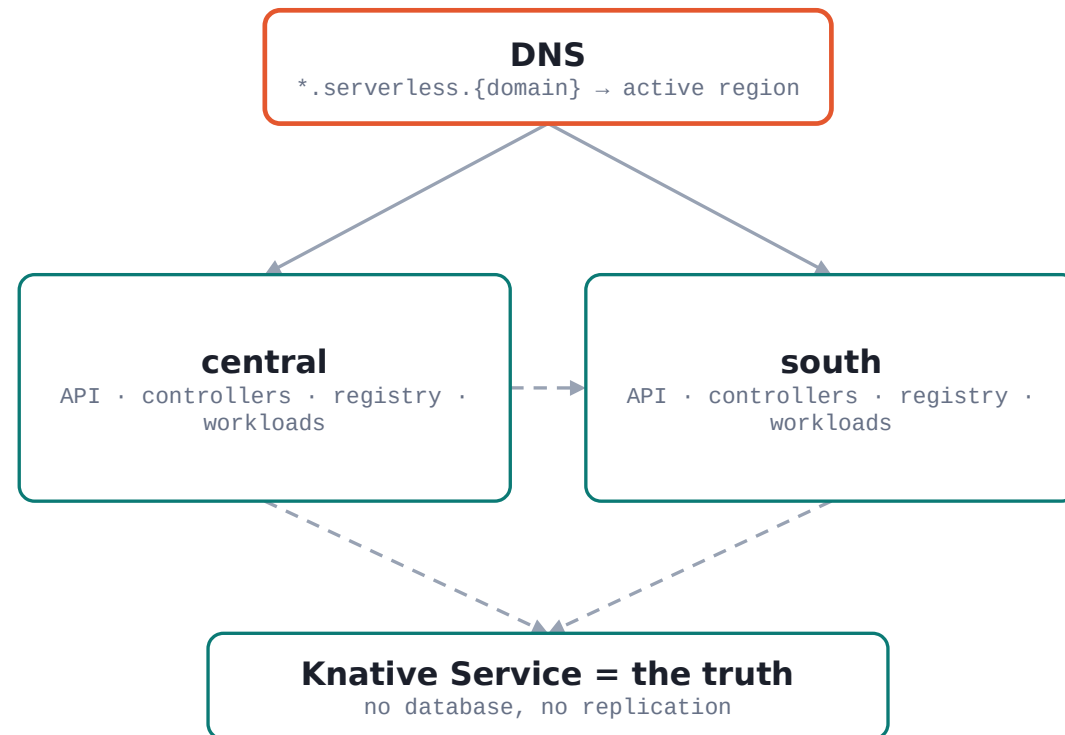
Talking to the cluster

- Client certificate mTLS, always
- No kubeconfig, no service-account token
- The certificate's CN is the user
- Address derived from the region name
- Every write is server-side apply



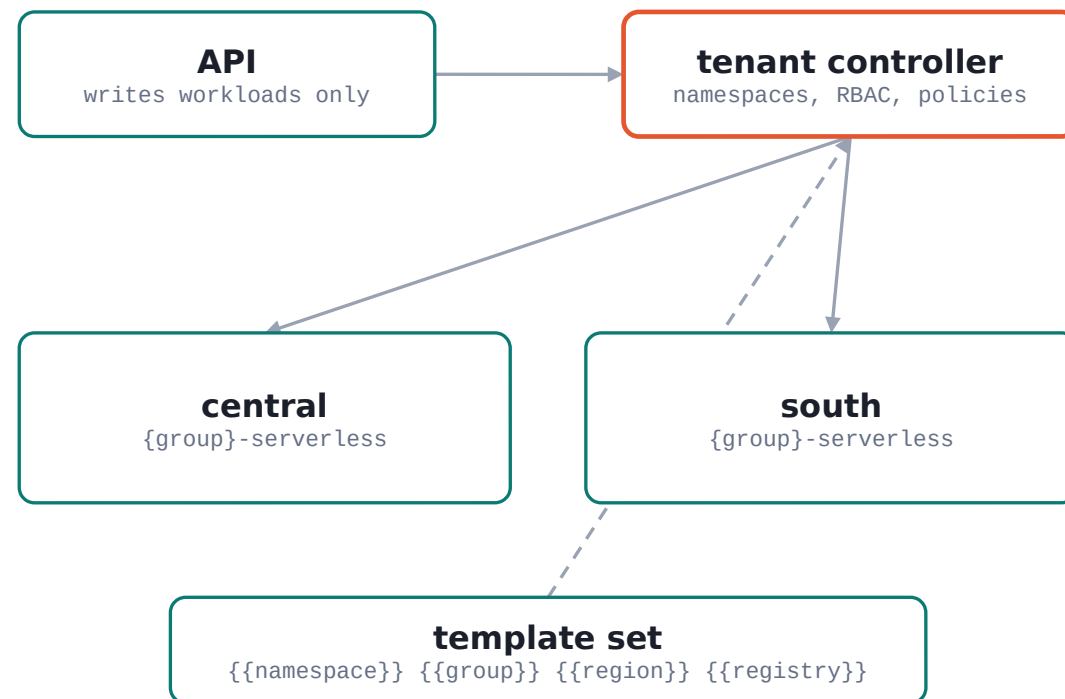
Active/active, two regions

- The API runs in both regions
- One deploy fans out to both
- No leader election, deterministic specs
- A region builds what it runs
- State lives in the Knative Service



The tenant controller

- A namespace per SSO group
- Separate process, separate certificate
- The API never creates namespaces
- Provision before every deploy, fail closed
- Reconcile the local cluster only
- Garbage-collect empty namespaces, opt-in



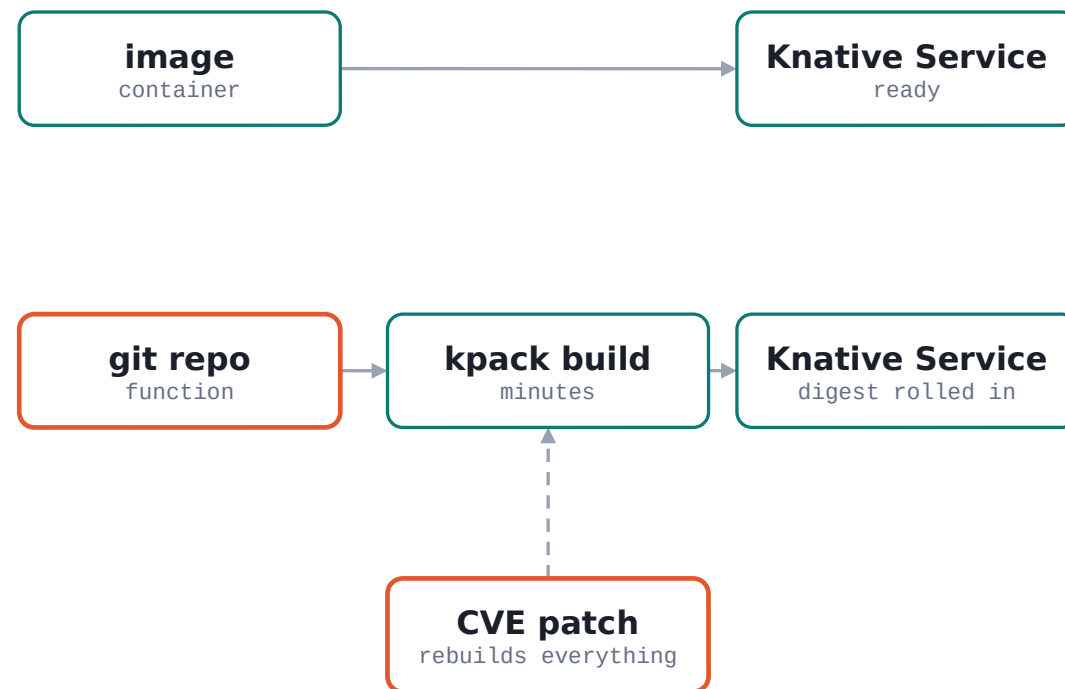
03

Functions & builds

'Here is my repo' hides a build.

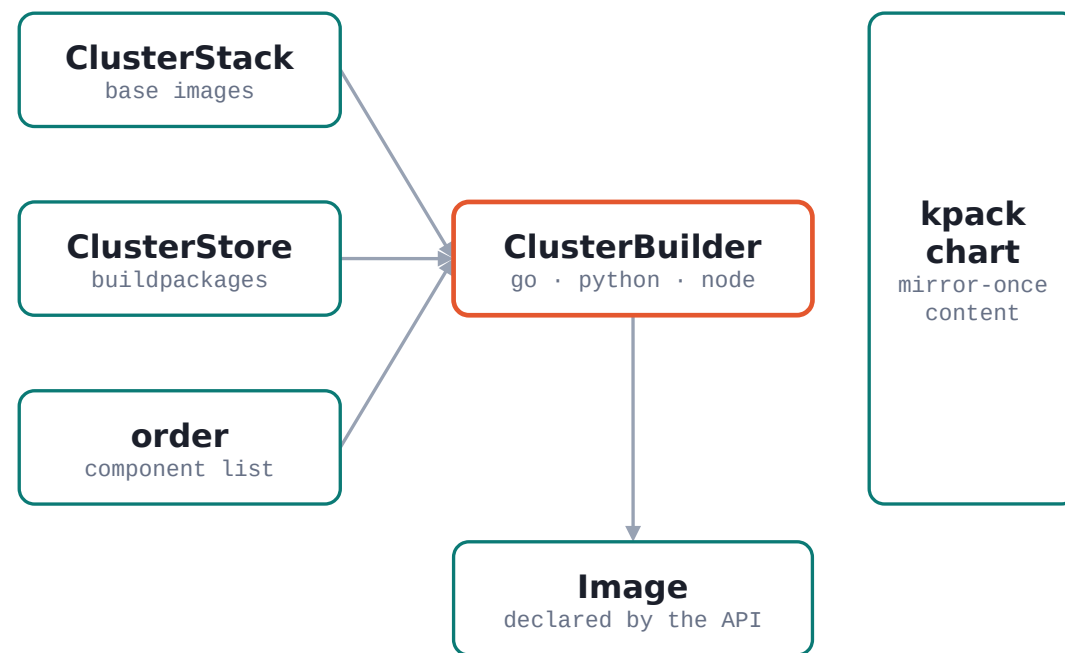
Why functions are different

- Container: run this image
- Function: here is my repository
- A build takes minutes, not milliseconds
- A build is retried and logged
- Rebuilds fire on CVE patches, unasked



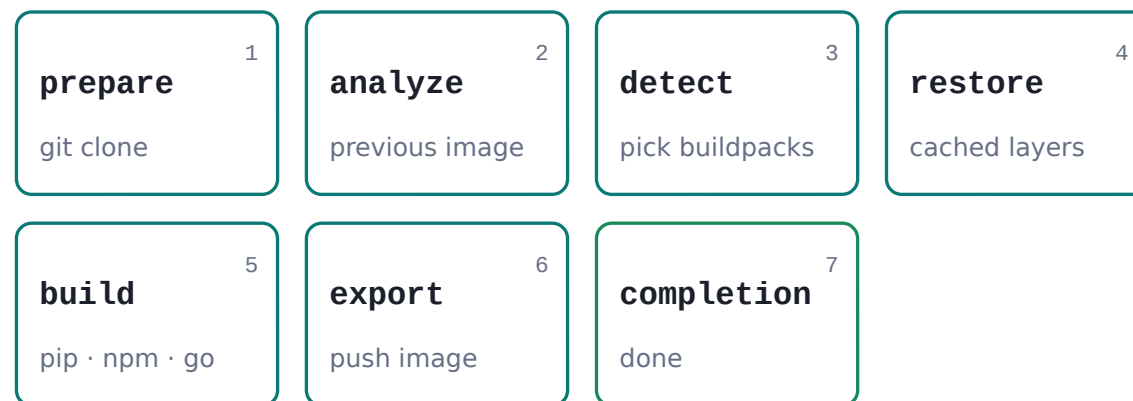
Buildpacks and kpack

- No Dockerfile, buildpacks detect the language
- Rebase: patch the base without rebuilding
- kpack: Image, Build, Pod
- Our kpack repo is a Helm chart
- Airgap mirror scripts included
- ClusterBuilders: go, python, node



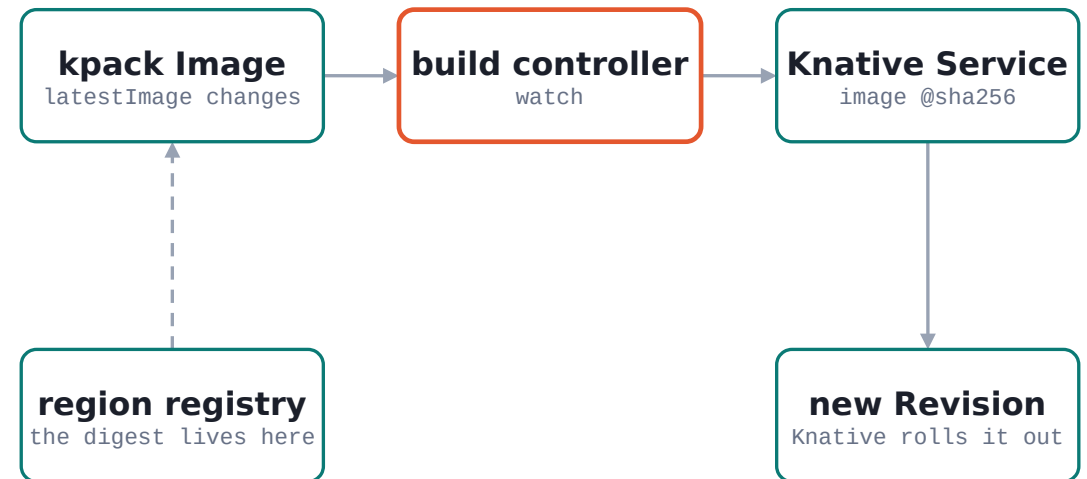
A build, step by step

- Image → SourceResolver → Build → Pod
- Each phase is its own container
- Each phase has its own log
- Pushed to this region's registry
- Five triggers; two need nobody



The build controller

- Watches Image.status.latestImage
- Applies the digest onto the Service
- Both ends stay in one region
- Only ever writes digests
- Ships with no web stack



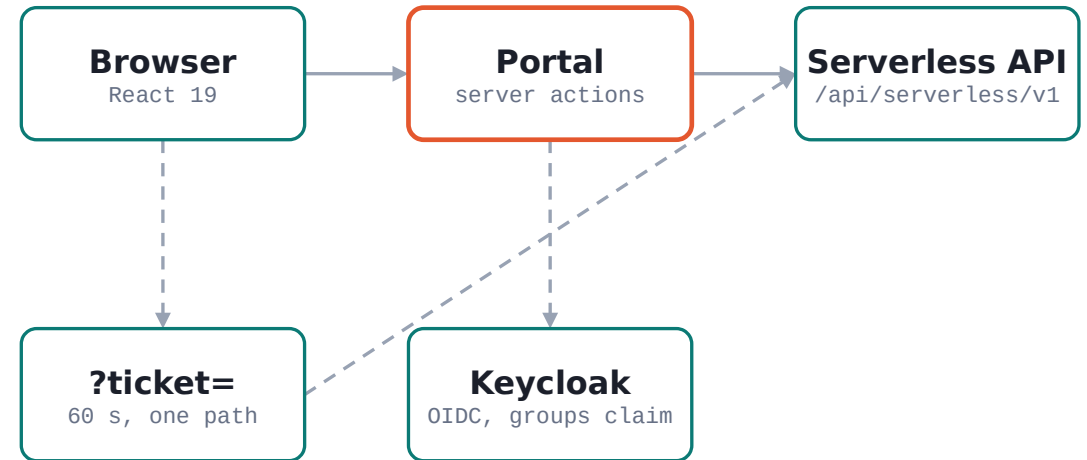
04

Portal & the future

A console, and a library for the next API.

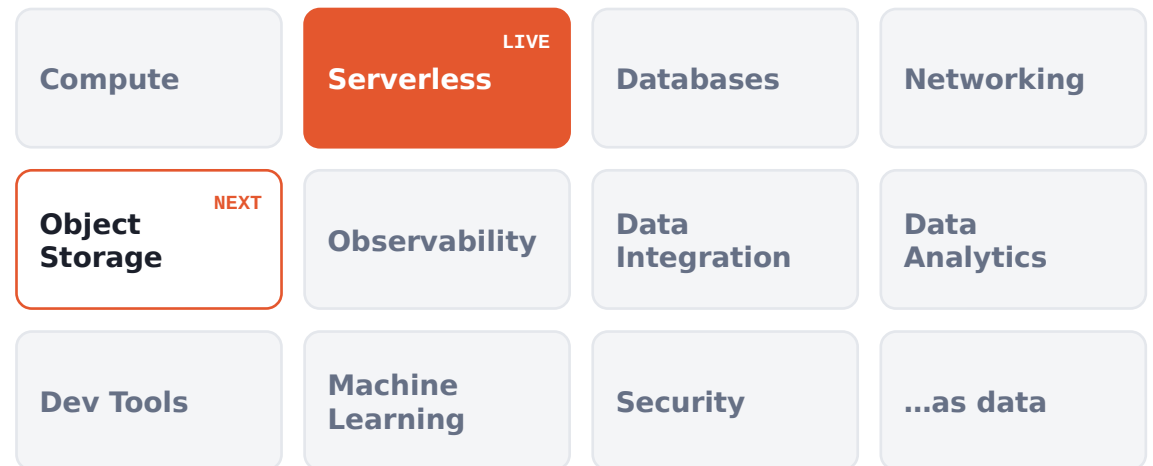
How the portal wires it

- Next.js 16, server components
- Keycloak token stays server-side
- The SSO group is the project
- Forms are built from /info
- Live logs over SSE tickets
- Airgap-native: no CDN, no fonts



A console, not a Serverless UI

- Serverless is live, Storage is next
- An offering is data, not code
- The shell is already generic
- Next: shared layer, generated client



cloudlet-apis: start at step five

- Extracted from the Serverless API
- core, web and auth extras
- Layering enforced by tests
- A new API wires in five steps
- Same group rules everywhere

[auth]

OIDC · admin key · token proxy ·
tickets

[web]

health · error envelope · offline docs

core

errors · names · logging · request id

Code in, a running service out.

4

repositories

3

services

23

endpoints

2

regions

3

runtimes

0

kubectl

- Operator owns Knative; we own the contract
- Deterministic specs, so no leader
- Builds declared, watched, rolled in
- The second product is cheaper

**Thank you.
Questions?**